From crude oil to consumer inflation: transmission and attenuation across India's price system

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Abstract

India imports more than four fifths of the crude it consumes, but the transmission of that external cost shock through domestic prices is uneven. This paper studies oil pass-through as a layered process rather than as a single inflation coefficient. It traces the shock across four price layers: headline wholesale inflation, retail petrol prices at the pump, fuel-sensitive wholesale and consumer components, and headline consumer inflation. The empirical core uses short-run asymmetric autoregressive distributed lag models with Newey–West inference on monthly data, with the headline WPI series chained back to 1983 and the fuel and power component to 1995. A complementary nonlinear ARDL battery in levels is reported in the appendix as supplementary long-run evidence. Three findings stand out. Pass-through into headline WPI is statistically detectable but small (cumulative pass-through of about 0.030 for positive oil shocks). It is much larger in retail petrol and in fuel-sensitive wholesale prices. It then attenuates sharply before reaching headline CPI, where the cumulative effect is not statistically distinguishable from zero. A pre/post-2010 split shows materially stronger headline pass-through in the post-deregulation period, consistent with a more market-linked pricing regime for transport fuels. Short-run asymmetry is tested but is not the paper's central empirical result. The main contribution is an integrated map of transmission and attenuation across India's price system, showing that domestic pricing institutions and index composition absorb much of the upstream oil shock before it reaches household inflation.

Keywords: oil price pass-through; imported inflation; India; wholesale inflation; consumer inflation; retail fuel pricing; deregulation; asymmetry.

1 Introduction

India is a structural net importer of crude oil. The country sourced close to 87 percent of its crude requirement from abroad in 2023–24, which makes the rupee value of a barrel of Brent one of the clearest external cost shocks entering the domestic price system in any given month (Petroleum Planning and Analysis Cell, 2024). The policy question that follows is straightforward but important: when crude prices move, how far does that movement travel through India's inflation structure, and how much of it reaches the prices households actually face?

Existing work on India has usually answered that question one index at a time. Some studies examine WPI in isolation (Bhoi et al., 2012; Suresh et al., 2026). Others focus on CPI subgroups or headline CPI (Abu-Bakar and Masih, 2018; Sek, 2019). Another strand studies retail fuel pricing after the June

2010 petrol deregulation and the October 2014 diesel deregulation (Pradeep, 2022; Deheri and Carmel, 2024). These studies are useful, but they leave a common gap. India's pricing system operates as a chain (international crude → exchange rate → domestic refined products → wholesale energy → consumer fuel → headline CPI), yet the literature rarely follows that chain end-to-end within a single empirical design. As a result, two findings that are both true are often discussed as if they were in tension: oil shocks pass through to wholesale inflation, and oil shocks barely move headline CPI. The present paper asks where, within the chain, the shock is amplified and where it weakens.

The paper addresses that question by estimating pass-through at four points along the price system. The headline WPI series is built by chaining four official base years (1981–82, 1993–94, 2004–05, 2011–12) into a single 528-observation monthly series running from April 1982 through March 2026, which provides the longest historically defensible wholesale inflation horizon currently available for India. The WPI Fuel and Power subindex is chained over three base years and runs from April 1994 through March 2026. The retail fuel layer is the PPAC monthly Delhi retail selling price for petrol from August 2004 onward. Headline CPI on the All-India 2010=100 base is the consumer endpoint. We also use the CPI Fuel and Light subgroup as a shorter bridge series. Because of the post-2011 CPI rebase, that bridge series is treated as supporting evidence rather than as one of the paper's headline mandates.

The empirical strategy is intentionally restrained. The main specifications are short-run asymmetric ADL models in log-differences. Each includes twelve own lags, lags zero to six on positive and negative oil-change components, month fixed effects, and Newey–West HAC standard errors. Cumulative pass-through is reported as the sum of the positive (CPT^+) and negative (CPT^-) coefficients, with Wald tests for each sum and for the symmetry restriction $\mathrm{CPT}^+ = \mathrm{CPT}^-$. A bootstrap symmetry test based on a restricted-residual circular block scheme provides finite-sample inference. A nonlinear ARDL battery in levels, estimated following Pesaran et al. (2001) and Shin et al. (2014), is reported in the appendix as supplementary long-run evidence rather than as the paper's core identification device.

The results support a clear layered reading. In the long historical sample, headline wholesale inflation responds to rupee oil shocks with $\mathrm{CPT}^+ = 0.030$ ($p = 0.024$) and $\mathrm{CPT}^- = 0.037$ ($p = 0.001$). These coefficients are small but precisely estimated, and asymmetry is not rejected (asymptotic $p = 0.673$, bootstrap $p = 0.746$). Pre/post-2010 splits show that most of the headline signal is concentrated in the post-deregulation period, where CPT^+ rises to 0.074 and CPT^- to 0.081. At the fuel-sensitive layer, the magnitudes change substantially. WPI Fuel and Power has $\mathrm{CPT}^+ = 0.287$ and $\mathrm{CPT}^- = 0.268$, both with $p < 0.001$. PPAC retail petrol responds even more strongly, with $\mathrm{CPT}^+ = 0.346$ and $\mathrm{CPT}^- = 0.191$ on a 245-month sample beginning in August 2004. By contrast, once the shock reaches headline CPI, the cumulative effect is very small: $\mathrm{CPT}^+ = 0.021$ with $p = 0.122$, while CPT^- is effectively zero ($p = 0.94$). In other words, much of the upstream movement is absorbed before it reaches the consumer aggregate.

The paper makes three contributions. First, it provides an integrated map of where the oil-price signal survives and where it attenuates, rather than treating WPI and CPI as competing answers to the same question. Second, the long historical wholesale sample anchors the main pass-through estimate in a 43-year window, which is unusual in the India literature, where many studies end in the 2010s. Third, the paper reframes the asymmetry question. Short-run asymmetry is tested, but it is not the most robust empirical feature of the data. The more stable result is attenuation across layers. The contribution is therefore one of integration and comparative mapping rather than methodological novelty, and the argument is framed accordingly.

The rest of the paper is organised as follows. Section 2 sets out the institutional background and

positions the paper within the literature. Section 3 describes the data and empirical strategy. Section 4 presents the historical wholesale results. Section 5 develops the mechanism and attenuation evidence across the four layers. Section 6 reports robustness checks and summarises the appendix NARDL results. Section 7 concludes with the main policy interpretation.

2 Institutional background and focused literature

Two institutional features shape the way oil shocks enter Indian inflation. The first is the foreign exchange channel. Brent is priced in dollars and India pays for oil imports in dollars, so the rupee cost of a barrel is the product of Brent and the INR/USD exchange rate. A five percent rupee depreciation against a flat Brent price therefore produces the same imported-cost effect as a five percent increase in Brent against a stable exchange rate. The second is the domestic pricing regime for refined products, which changed substantially over the sample period. From the 1970s into the early 2000s, India operated an Administered Pricing Mechanism under which the union government set retail prices for petrol, diesel, kerosene and LPG and absorbed gaps between domestic and import-parity prices through subsidies and oil bonds. That framework was dismantled gradually. Petrol was deregulated in June 2010 and diesel in October 2014, while kerosene and LPG remained partially subsidised. At the same time, tax components (central excise and state VAT) came to account for a large share of the pump price, giving the government a meaningful buffer against international price movements.

Taken together, these features imply a layered transmission structure. In a more market-linked retail fuel regime, an oil shock should pass relatively quickly into petrol and diesel prices. From there it should show up in fuel-sensitive components of the wholesale and consumer indices. Whether it reaches headline inflation depends on the fuel weight in the relevant basket and on whether non-fuel components move in the same direction. Under a more administered regime, the same shock should be absorbed more heavily at the retail stage through subsidy adjustments and tax changes. The pre- and post-2010 subsamples therefore provide a useful institutional comparison for whether deregulation made the pass-through chain more permissive.

The Indian literature on oil pass-through offers partial answers to this institutional setup. Early work using linear VAR and cointegration methods on WPI generally reported small but statistically significant pass-through coefficients (Bhoi et al., 2012; Bhanumurthy et al., 2012). A later strand moved to nonlinear ARDL models to test for asymmetry. Abu-Bakar and Masih (2018) report long-run asymmetric pass-through to Indian inflation using NARDL. Chattopadhyay and Mitra (2015) and Pal and Mitra (2016) study asymmetry in oil product prices and find that it is more visible where market-linked pricing actually operates. Pandey and Shettigar (2019) document partial pass-through from crude to retail fuel in the early post-deregulation period. Pradeep (2022) focuses on the diesel reform and reports stronger asymmetric pass-through to disaggregated wholesale and consumer prices after reform. Deheri and Carmel (2024) also document asymmetric pass-through from crude to retail fuel. Two further studies are especially relevant for the present design. Suresh et al. (2026) estimate a NARDL system for Indian wholesale prices and find long-run asymmetry with weaker short-run asymmetry, which broadly matches the supplementary appendix evidence reported here. Sek (2019) provide cross-country panel evidence that the WPI–CPI gap in pass-through is wider in oil-importing economies, offering a useful comparative benchmark.

What the literature still lacks is a single empirical architecture that traces the full chain from crude to headline CPI using consistent monthly data. Most studies choose one estimator family (NARDL,

SVAR, or VECM), one inflation measure, and one part of the transmission process. The contrast between headline and fuel-sensitive indices is sometimes mentioned, but attenuation across layers is rarely the central object of inference. Sadath and Acharya (2021), using an SVAR, comes closest to a layered macro framing, although that paper is more concerned with output responses than with pricing institutions. The present paper's contribution is therefore more architectural than methodological. It keeps the short-run asymmetric ADL as the main workhorse, places NARDL in the appendix, and treats the gap between pass-through estimates across layers as the substantive result.

A brief methodological clarification is useful here. Parts of the Indian pass-through literature have treated headline WPI as a proxy for input-cost inflation and headline CPI as a proxy for household-facing inflation, then argued over which is the more relevant policy measure. This paper does not frame the problem that way. WPI and CPI answer different questions. WPI is more informative about upstream cost transmission and imported input-price pressures, whereas CPI is more informative about household-facing inflation. Studying both jointly allows the paper to distinguish where oil shocks enter the price system from whether they survive all the way to headline consumer inflation. The post-2014 monetary policy framework that targets headline CPI (Reserve Bank of India, 2014) makes the CPI endpoint essential for policy interpretation, but it does not make the wholesale evidence redundant. On the contrary, wholesale evidence becomes more informative once we know that the upstream signal does not automatically carry through to the target index.

3 Data and empirical strategy

The dataset is monthly. Brent crude in US dollars per barrel comes from the World Bank Commodity Markets Pink Sheet. The INR/USD spot rate comes from FRED (series `EXINUS`). The rupee oil price is constructed as the product of Brent and EXR. Wholesale prices come from the Office of the Economic Adviser (OEA) under the Department for Promotion of Industry and Internal Trade. Headline WPI and the WPI Fuel and Power subindex are constructed by chain-linking the four official OEA base series (1981–82, 1993–94, 2004–05 and 2011–12) using the official linking factors, with all series rebased to 2011–12 = 100. This gives a 528-observation headline series running April 1982 through March 2026 and a 384-observation Fuel and Power series running April 1994 through March 2026. The retail fuel layer is the PPAC monthly Delhi retail selling price for petrol, available from August 2004. The consumer endpoint is the All-India headline CPI series. The CPI Fuel and Light subgroup is taken from the post-2011 base and used as a supporting bridge series rather than as a primary index. After inner joins on common dates, the active estimation samples cover roughly 43 years for headline WPI, 31 years for WPI Fuel and Power, and 20 years for the PPAC and CPI specifications. Table 1 summarises the series.

All series are converted to month-on-month log differences. Let $\Delta y_t = \ln Y_t - \ln Y_{t-1}$ and $\Delta x_t = \ln X_t - \ln X_{t-1}$ where X_t denotes the relevant oil measure. The asymmetric oil change is decomposed into a positive and a negative part:

$$\Delta x_t^+ = \max(\Delta x_t, 0), \qquad \Delta x_t^- = \min(\Delta x_t, 0). \tag{1}$$

The main estimating equation is the short-run asymmetric ADL

$$\Delta y_t = \alpha + \sum_{i=1}^{12} \phi_i \Delta y_{t-i} + \sum_{j=0}^{6} \beta_j^+ \Delta x_{t-j}^+ + \sum_{j=0}^{6} \beta_j^- \Delta x_{t-j}^- + \gamma' Z_t + \mu_m + u_t, \tag{2}$$

Table 1: Series, sources, transformations, and sample spans.

Series	Source	Transform.	Active sample
Brent crude (USD/bbl)	World Bank Pink Sheet	$\Delta\ln$	1983-05 to 2026-03
INR/USD spot rate	FRED `EXINUS`	$\Delta\ln$	1983-05 to 2026-03
Rupee oil price	Brent $\times$ EXR (constructed)	$\Delta\ln$	1983-05 to 2026-03
Headline WPI (2011-12 = 100)	OEA, chained across four bases	$\Delta\ln$	1983-05 to 2026-03
WPI Fuel & Power (2011-12 = 100)	OEA, chained across three bases	$\Delta\ln$	1995-05 to 2026-03
PPAC retail petrol, Delhi	Petroleum Planning and Analysis Cell	$\Delta\ln$	2004-08 to 2024-12
Headline CPI (All-India, 2010 = 100)	MoSPI / FRED `INDCPIALLMINMEI`	$\Delta\ln$	2004-08 to 2024-12
CPI Fuel & Light (post-2011 base)	MoSPI	$\Delta\ln$	2011-05 to 2024-12

Notes: Wholesale series chained to a common 2011–12 = 100 base using OEA linking factors. Active samples reported after inner joins on common dates. The CPI Fuel & Light series begins in May 2011 because the post-2011 base is the longest harmonised consumer fuel index currently available.

where Z_t collects controls (notably $\Delta\ln\mathrm{EXR}_t$ in the Brent decomposition), μ_m are calendar-month fixed effects, and u_t is an error process whose autocorrelation we accommodate through Newey–West standard errors with a Bartlett kernel and Newey–West lag truncation. The cumulative pass-through summaries are

$$\mathrm{CPT}^{+} = \sum_{j=0}^{6} \beta_j^{+}, \qquad \mathrm{CPT}^{-} = \sum_{j=0}^{6} \beta_j^{-}, \tag{3}$$

with significance assessed using HAC Wald tests on the linear restrictions $\mathrm{CPT}^{+} = 0$, $\mathrm{CPT}^{-} = 0$ and the symmetry restriction $\mathrm{CPT}^{+} = \mathrm{CPT}^{-}$. For the symmetry test we also report a finite-sample p-value from a restricted-residual circular block bootstrap with 4,999 replications.

The choice of an ADL in differences rather than a NARDL in levels is deliberate. Standard unit-root tests reject I(1) for all the differenced series we use; ADF and Phillips–Perron statistics on $\Delta\ln\mathrm{WPI}$ are -12.22 and -14.39. KPSS gives a borderline rejection of stationarity for $\Delta\ln\mathrm{WPI}$ ($\hat{\eta} = 0.90$ against a 5% critical value of 0.46), which we read as a known KPSS over-rejection in long samples with structural breaks rather than as evidence of a unit root in the differences (see Reserve Bank of India, 2014). A Bai–Perron break test finds one break in $\Delta\ln\mathrm{WPI}$ at September 2013, which is the disinflationary regime change familiar from the inflation-targeting transition. With this in mind, ADL in differences with Newey–West inference is the cleaner short-run vehicle. We retain a NARDL-in-levels battery in the appendix for two reasons: it provides a long-run cointegration check, and it is the dominant method in the Indian pass-through literature, so silence on it would be conspicuous.

The retail-fuel mechanism model uses $\Delta\ln\mathrm{Petrol}_t^{\mathrm{Delhi}}$ as the dependent variable and asymmetric Brent (in USD) as the regressor, with $q = 3$ lags on each side. Although the petrol pricing regime moved to fortnightly in 2014 and to daily revisions in 2017, monthly averaging is appropriate for the cross-layer comparison and avoids the Brent–EXR averaging mismatch that a daily specification would inherit. The CPI bridge model regresses $\Delta\ln\mathrm{Fuel\&Light}_t^{\mathrm{CPI}}$ on asymmetric retail petrol changes; this is reported as a bridge equation only because the available CPI Fuel and Light series begins in May 2011 and falls below the 20-year window we treat as the minimum sample for headline mandates.

Granger predictive precedence is reported alongside the ADL evidence as a complementary check. We treat Granger statistics as evidence of lead–lag predictability rather than of structural causality, in line

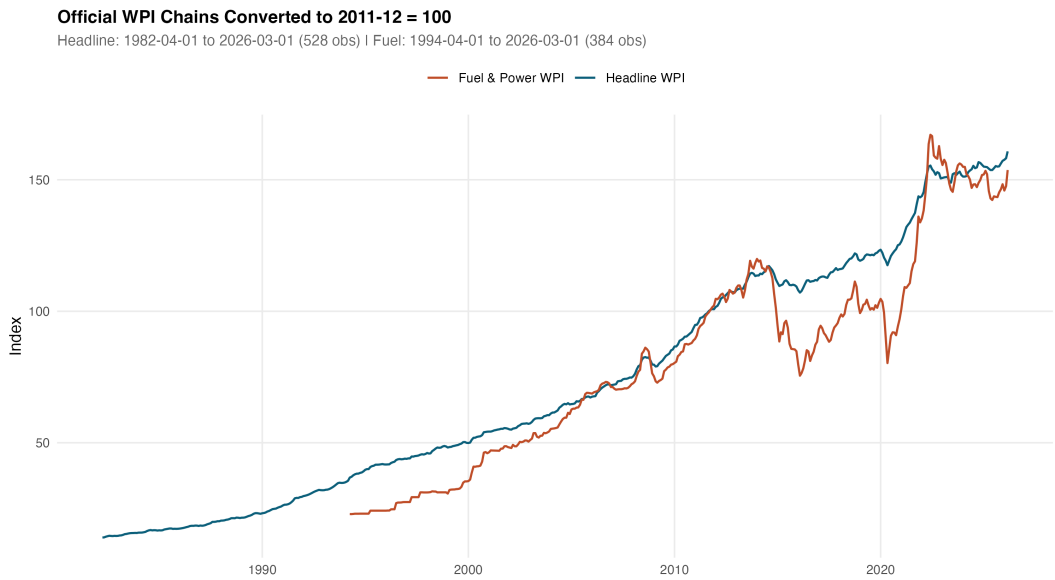

Figure 1: Chained WPI series, 2011-12 = 100. Headline WPI runs from April 1982 (528 monthly observations); Fuel & Power begins April 1994 (384 observations). Splice points across the four official OEA base years are marked.

Table 2: Headline WPI ADL on rupee oil. Cumulative pass-through over lags 0–6.

Specification	N	Span (yrs)	Adj. R^2	CPT^+	CPT^-	Asym. p
Headline WPI ADL, INR oil	515	42.92	0.421	0.030	0.037	0.673
p-value on cumulative coefficient				0.024	0.001	
Bootstrap symmetry ($B = 4{,}999$, restricted-residual circular block):						
Bootstrap p on $\text{CPT}^+ = \text{CPT}^-$						0.746

Notes: Sample 1983-05 to 2026-03. Specification: $\Delta \ln \text{WPI}_t$ on twelve own lags, lags 0–6 of $\Delta \ln \text{Oil}^{\text{INR}+}_t$ and $\Delta \ln \text{Oil}^{\text{INR}-}_t$, and calendar-month fixed effects. Newey–West HAC standard errors. Cumulative pass-through $\text{CPT}^{\pm}$ is the sum of the corresponding seven coefficients. Asymmetry p is the HAC Wald test of $\text{CPT}^+ = \text{CPT}^-$.

with the conservative reading common in the recent literature.

4 Historical wholesale pass-through

We start with the longest sample. Table 2 reports the headline WPI ADL on rupee oil for May 1983 through March 2026 ($N = 515$, span 42.92 years).

The picture is consistent across the two cumulative summaries. Positive oil shocks raise wholesale inflation by 3.0 percent of the shock over the seven-month window, with $p = 0.024$. Negative shocks subtract 3.7 percent, with $p = 0.001$. Both numbers are small but precise. The Wald symmetry test on the seven-lag restriction yields an asymptotic p-value of 0.673 and a bootstrap p-value of 0.746; the data give no support to short-run asymmetry over the full historical sample. The Brent + EXR decomposition (reported in the next section as a robustness check) returns a near-identical CPT^+ of 0.031 and CPT^- of 0.037, which suggests that the rupee oil aggregate is not hiding a sign-cancelling combination of price and exchange-rate movements. Adjusted R^2 for the headline ADL is 0.42. A Breusch–Godfrey test at twelve lags returns $p = 0.84$. The HAC RESET test passes ($p = 0.35$). The recursive CUSUM test passes at conventional levels ($p = 0.39$).

The interesting result is the subsample split. Indian petrol prices were deregulated in June 2010 and

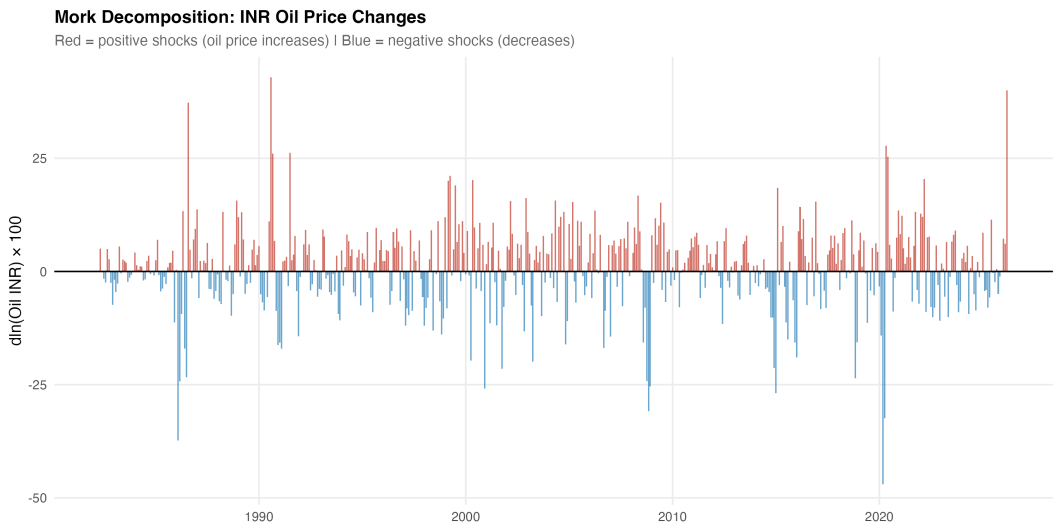

Figure 2: Decomposition of the rupee oil price into Brent (USD) and the INR/USD rate. Plotted in log levels with each series indexed to its January 2000 value. The two cost-side drivers move out of phase in some windows (notably 2003–2008 and 2014–2016), which is the variation the Brent + EXR robustness specification in Section 6 exploits.

Table 3: Pre/post-2010 subsample comparison. Headline WPI and Fuel & Power ADL.

Model	Subsample	N	CPT^+	p	CPT^-	p
Headline WPI ADL	Pre-2010 (1983-05 to 2010-03)	323	0.012	0.381	0.025	0.043
Headline WPI ADL	Post-2010 (2010-04 to 2026-03)	192	0.074	0.004	0.081	0.000
Fuel & Power ADL	Pre-2010 (1995-05 to 2010-03)	179	0.092	0.168	0.238	0.000
Fuel & Power ADL	Post-2010 (2010-04 to 2026-03)	192	0.524	0.000	0.438	0.000

Notes: ADL specification as in Table 2. Split point April 2010, the month after India's June 2010 petrol deregulation. The post-2010 window also contains the October 2014 diesel deregulation. The split should be read as descriptive of the post-deregulation regime and not as a causal identification of reform effects.

diesel in October 2014. We split the sample at April 2010 to run a clean comparison and re-estimate the same ADL on the two windows.

The pre-2010 estimates are weak. Positive shocks pass through with $\text{CPT}^+ = 0.012$ ($p = 0.38$) and negative shocks with $\text{CPT}^- = 0.025$ ($p = 0.04$). After deregulation, the same model on the post-2010 sample returns $\text{CPT}^+ = 0.074$ ($p = 0.004$) and $\text{CPT}^- = 0.081$ ($p < 0.001$). The post-period magnitudes are roughly six times the pre-period magnitudes on the positive side and three times on the negative side. The fuel-and-power subindex tells the same story with larger magnitudes. Pre-2010 fuel CPT^+ is 0.092 and not significant; post-2010 it jumps to 0.524 and is highly significant. We read these comparisons as broadly consistent with the institutional shift toward more market-linked pricing, but we are deliberate about not calling them a causal identification of reform effects. The 2010–2014 window contains both monetary regime changes and global oil cycles, and the split-sample comparison cannot net these out cleanly.

Two stylised observations help frame what follows. First, even at its full-sample maximum, headline WPI pass-through is small in absolute terms: a one percent rupee oil rise lifts wholesale inflation by about three basis points cumulatively over seven months. Second, the absence of short-run asymmetry in the headline series sits in tension with the asymmetric findings reported by parts of the Indian NARDL literature (Abu-Bakar and Masih, 2018; Suresh et al., 2026). We pick that tension up explicitly in Section 6, where the appendix NARDL battery does detect long-run asymmetry while the short-run ADL does

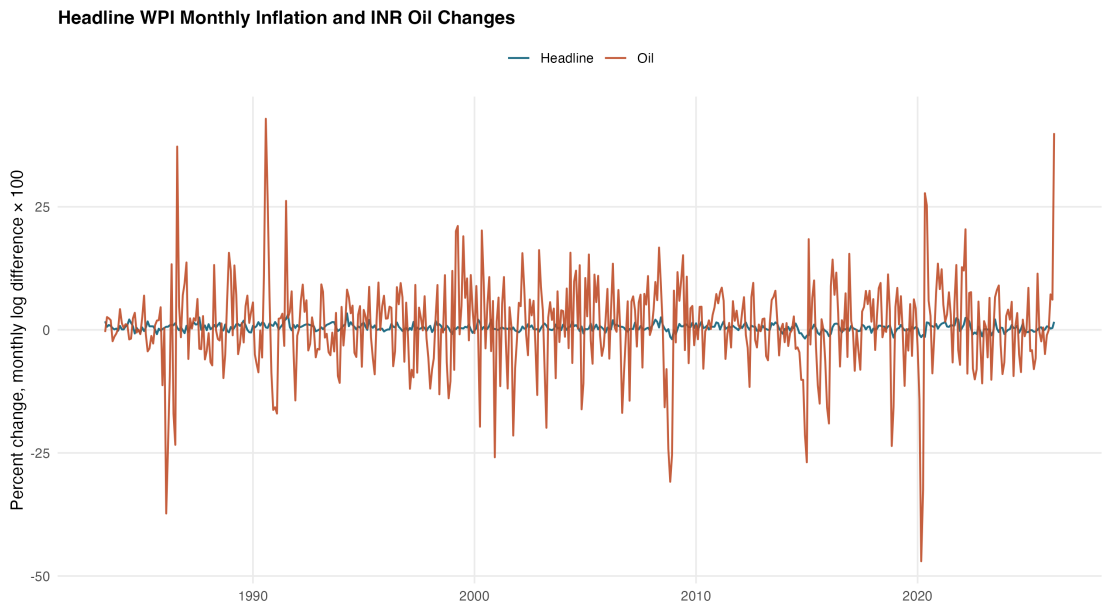

Figure 3: Headline WPI inflation and the rupee oil change, monthly, 1983–2026. Both series are plotted in $\Delta \ln$ form. The wholesale series tracks the rupee oil change visibly through the 1990s and from 2010 onward, with a quieter middle period that the subsample split picks up.

not. The two methods are not contradictory. They are pricing different things. ADL in differences is asking whether the seven-month sum of positive coefficients matches the seven-month sum of negative coefficients. NARDL in levels is asking whether the long-run multipliers from positive and negative cumulative oil shocks differ. Long-run asymmetry can coexist with short-run symmetry. Our preference for the short-run reading in the body of the paper is partly a matter of inferential transparency. The bootstrap distribution for the symmetry test is well-behaved on the differenced specifications, and the standard errors do not require an assumed cointegration structure to be valid.

The long-horizon wholesale evidence, taken together, supports a moderate but credible historical pass-through. It also points to where the action sits in the more recent data. The next section turns to the layers in which the same shock shows up an order of magnitude larger.

5 Mechanism and attenuation across layers

The point of the integrated design is in this section. We move along the chain from crude to consumer, and at each step we ask how much of the upstream signal is preserved.

Figure 6 sketches the chain we estimate. Brent and the rupee/dollar rate combine into the rupee oil price. The rupee oil price feeds into PPAC retail petrol at the pump. Retail fuel prices feed into the fuel-sensitive subindices of WPI (Fuel and Power) and CPI (Fuel and Light). The fuel-sensitive subindices, together with non-fuel cost pressures, feed into the two headline aggregates. The arrows in the diagram are weighted by the cumulative positive pass-through estimates from our ADL specifications. The CPI endpoint arrow is dashed and thinned to convey what the data say: by the time the shock reaches headline CPI, very little of it is left.

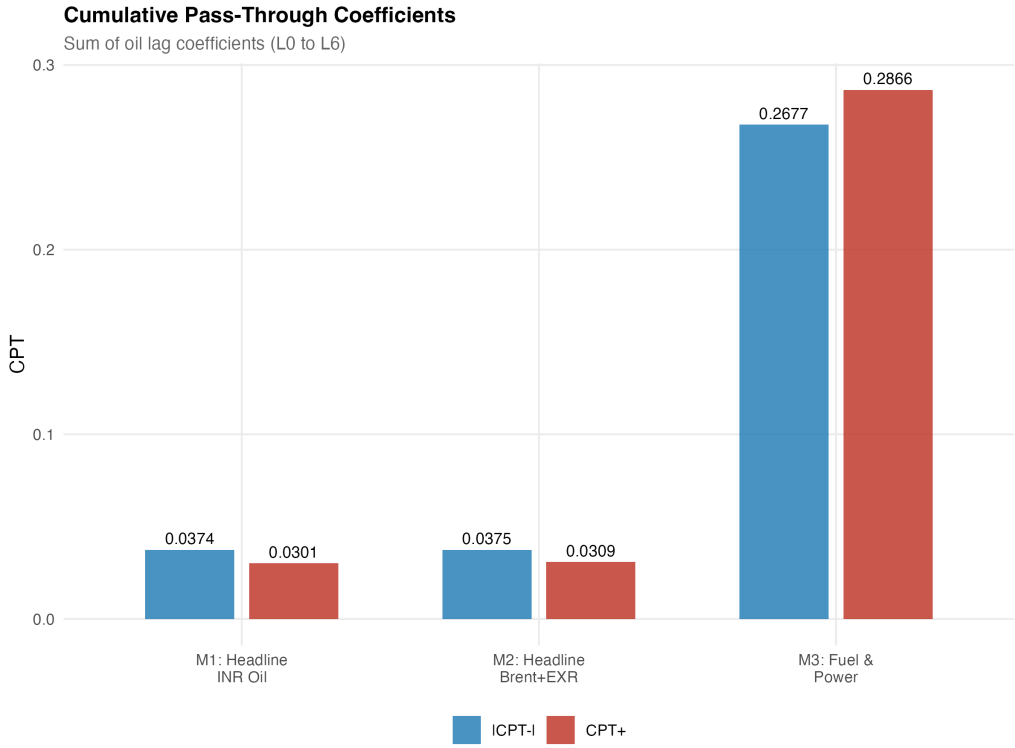

Figure 4: Cumulative pass-through profile for the headline WPI ADL. Lag-by-lag accumulation of the positive and negative rupee-oil coefficients from lag 0 through lag 6, with HAC bands. Most of the response is loaded onto the first three lags; the seven-lag sum stabilises at the values reported in Table 2.

5.1 Retail fuel and the wholesale fuel layer

Table 4 reports the two upstream mechanism models. Panel A is the PPAC Delhi retail petrol model on August 2004 to December 2024 ($N = 245$). Panel B is the WPI Fuel and Power ADL on May 1995 to March 2026 ($N = 371$).

The retail petrol response is the largest in the chain. $\text{CPT}^+ = 0.346$ ($p < 0.001$) and $\text{CPT}^- = 0.191$ ($p < 0.001$). Asymmetry is borderline ($p = 0.10$), with the asymmetry running in the conventional rockets-and-feathers direction: positive shocks are passed through faster and more fully than negative ones. This is the empirical signature most consistent with the descriptive picture of post-deregulation pump pricing in India. It is also broadly consistent with the magnitudes reported by Deheri and Carmel (2024) and Pradeep (2022) for the petrol channel.

The fuel-and-power subindex of WPI sits one step downstream and shows comparable but slightly smaller magnitudes, with $\text{CPT}^+ = 0.287$ and $\text{CPT}^- = 0.268$ (both $p < 0.001$). Symmetry is not rejected here ($p = 0.78$), which we read as a sign that the wholesale fuel index aggregates retail and bulk pricing channels in a way that washes out the asymmetry visible at the pump. We flag one diagnostic caveat: HAC RESET fails ($p < 0.001$) on the fuel and power model, which is most likely picking up a non-linearity in the level relationship that the differenced specification cannot remove. We retain the model in the main results because the cumulative pass-through estimates are stable across alternative lag choices and across COVID-window sensitivity checks (reported in Section 6), but we treat the fuel and power equation as the more fragile of the two upstream models.

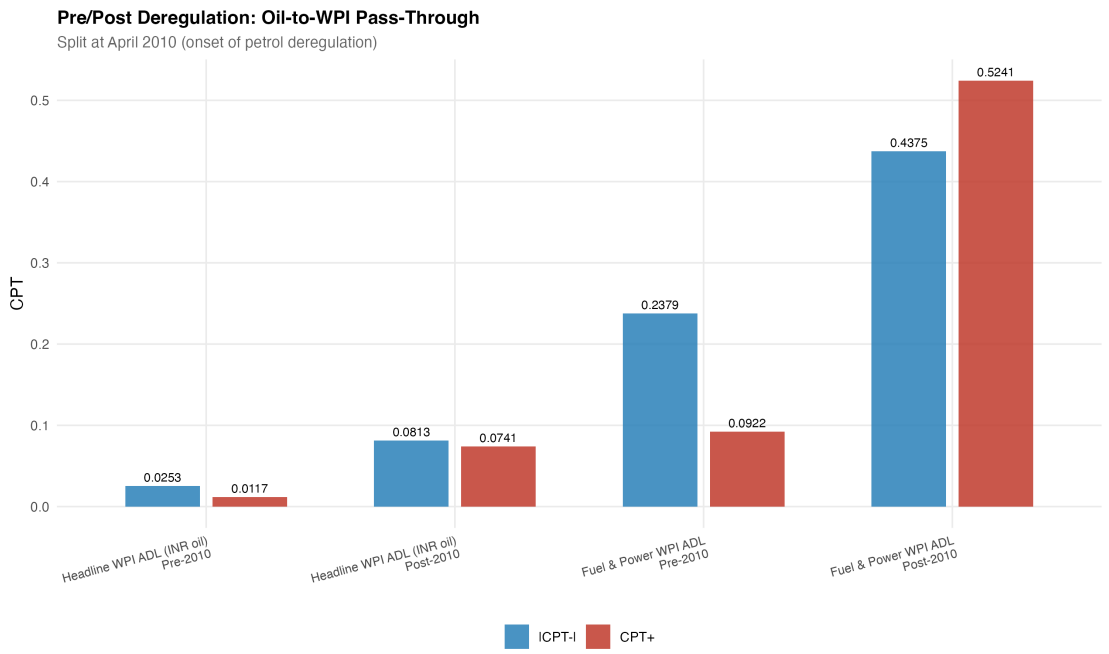

Figure 5: Pre/post-2010 subsample comparison of cumulative pass-through. Headline WPI and Fuel & Power ADL specifications, plotted as cumulative coefficient response over lags 0–6. The post-2010 magnitudes are visibly larger across both panels, broadly consistent with the institutional shift toward market-linked transport-fuel pricing.

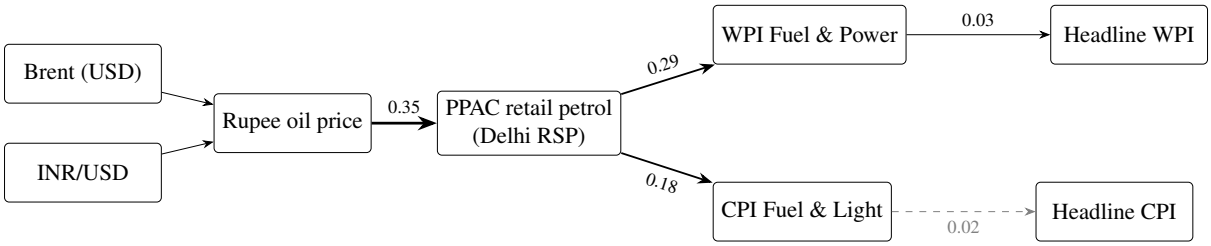

Figure 6: Conceptual transmission map. Edge labels are cumulative positive pass-through estimates (CPT^+) from the ADL specifications reported in Tables 2–5. The headline CPI arrow is dashed to mark that the corresponding coefficient is not statistically distinguishable from zero.

5.2 The consumer endpoint

Now move to the bottom of the chain. Table 5 reports the headline CPI ADL.

The headline CPI estimates are the smallest and the least precise in the chain. Positive shocks have $\text{CPT}^+ = 0.021$ with $p = 0.12$. Negative shocks are essentially flat: $\text{CPT}^- = 0.001$, $p = 0.94$. Symmetry is not rejected ($p = 0.24$). The bootstrap symmetry p-value is similar at 0.50. Two readings are possible. One is that headline CPI in India is genuinely insulated from oil shocks because the CPI basket is heavily food-weighted, with combined food and beverage weights close to half of the index. The other is that the consumer fuel and tax buffers absorb much of what does come through. Both are likely true, and the data do not let us cleanly separate them.

What the data do let us do is record one institutional fact that any policy reading should not lose. The October 2014 diesel-deregulation dummy in the headline CPI ADL is large and significant in the negative direction (estimate -0.339, $p < 0.001$). The October 2014 step is, of course, one event, and a single dummy cannot identify a structural channel. But the sign and magnitude are consistent with the institutional reading that diesel-tax revisions have at times moved in a direction that offsets, rather than reinforces, the international price signal in the consumer index.

Table 4: Mechanism layer ADL models. Retail petrol and wholesale fuel.

Specification	N	Span (yrs)	Adj. R^2	CPT^+	CPT^-	Asym. p
Panel A. PPAC Delhi retail petrol on Brent (USD), $q = 3$:						
$\Delta \ln \text{Petrol}_t$ ADL	245	20.42	0.236	0.346	0.191	0.100
p-value on cumulative coefficient				0.000	0.000	
Panel B. WPI Fuel & Power on rupee oil:						
$\Delta \ln \text{Fuel\&Power}_t$ ADL	371	30.92	0.463	0.287	0.268	0.783
p-value on cumulative coefficient				0.000	0.000	

Notes: Panel A sample 2004-08 to 2024-12. Panel B sample 1995-05 to 2026-03. Both panels report HAC (Newey–West) inference on the cumulative pass-through. Panel B includes a post-October-2014 diesel deregulation dummy and an April–September 2020 COVID dummy in addition to month fixed effects. The HAC RESET test fails on Panel B at $p < 0.001$ (see Section 6); other diagnostics pass.

Table 5: Headline CPI ADL on rupee oil. Consumer endpoint.

Specification	N	Span (yrs)	Adj. R^2	CPT^+	CPT^-	Asym. p
Headline CPI ADL, INR oil	245	20.42	0.449	0.021	0.001	0.241
p-value on cumulative coefficient				0.122	0.938	
Selected covariate effects (HAC t-tests on the levels coefficients):						
Diesel deregulation step (October 2014)		$\hat{\beta} = -0.339$		$p < 0.001$		
Petrol deregulation step (June 2010)		$\hat{\beta} = +0.116$		$p = 0.307$		
COVID window (April–September 2020)		$\hat{\beta} = -0.345$		$p = 0.507$		

Notes: Sample 2004-08 to 2024-12. Specification analogous to Table 2 with $q = 3$ on the asymmetric oil components, an IIP control for industrial activity, and step dummies for the petrol and diesel deregulation events. The bootstrap p-value on the symmetry restriction is 0.500.

5.3 Putting it together

Table 6 stitches the four CPT^+ estimates into a single layered comparison and adds the pre/post-2010 split for the two wholesale models.

The picture is monotone in the obvious direction. Retail petrol has $\text{CPT}^+ = 0.35$. The two fuel-sensitive subindices come in at 0.29 (WPI Fuel and Power) and 0.18 (CPI Fuel and Light). Headline WPI is at 0.03. Headline CPI is at 0.02 and not significant. Roughly an order of magnitude is lost between the pump and the consumer. Most of that loss happens between the fuel-sensitive subindices and the headline aggregates, which is mechanical: fuel weights in the headline indices are small (about 13 percent in the WPI 2011–12 series and about 7 percent in the CPI 2010 base), so a large move in fuel cannot mathematically produce more than a small move in the aggregate unless non-fuel components correlate with it. The CPI Fuel and Light bridge equation, although it sits on a shorter sample (164 monthly observations beginning May 2011 and therefore not eligible as a headline mandate), confirms the qualitative direction: PPAC petrol passes into CPI Fuel and Light with $\text{CPT}^+ = 0.178$ ($p = 0.002$) and a smaller negative-side response.

We pick out one judgement call. The evidence here is more cautious about asymmetry than the recent Indian NARDL literature, which often headlines asymmetric long-run multipliers as the primary result. We do find asymmetry signals, but they are concentrated at the pump, are borderline at conventional significance ($p = 0.10$), and are not robust to moving up the chain. The integrated reading is that the most stable and quantitatively meaningful empirical feature in this dataset is attenuation across layers, not asymmetry within any single layer.

Table 6: Integrated attenuation across price layers and pre/post-2010 split.

Layer	Dependent variable	N	Span	CPT^+	CPT^+ p
Panel A. Layered cross-section (full available samples):					
Retail fuel pump	$\Delta \ln \text{Petrol}_t^{\text{Delhi}}$	245	20.4 yrs	0.346	0.000
Wholesale fuel	$\Delta \ln \text{WPI Fuel\&Power}_t$	371	30.9 yrs	0.287	0.000
Consumer fuel (bridge)	$\Delta \ln \text{CPI Fuel\&Light}_t$	164	13.7 yrs	0.178	0.002
Wholesale headline	$\Delta \ln \text{WPI}_t$	515	42.9 yrs	0.030	0.024
Consumer headline	$\Delta \ln \text{CPI}_t$	245	20.4 yrs	0.021	0.122
Panel B. Pre/post-2010 split (wholesale models):					
Headline WPI, pre-2010	$\Delta \ln \text{WPI}_t$	323	1983–2010	0.012	0.381
Headline WPI, post-2010	$\Delta \ln \text{WPI}_t$	192	2010–2026	0.074	0.004
Fuel & Power, pre-2010	$\Delta \ln \text{Fuel\&Power}_t$	179	1995–2010	0.092	0.168
Fuel & Power, post-2010	$\Delta \ln \text{Fuel\&Power}_t$	192	2010–2026	0.524	0.000

Notes: All entries are cumulative pass-through estimates from short-run asymmetric ADL specifications, summed over lags 0–q on the relevant positive component. Panel A uses each layer's full active sample. The CPI Fuel & Light bridge equation is shown for completeness only; its sample begins in 2011 and falls below the 20-year window we treat as the minimum for headline mandates.

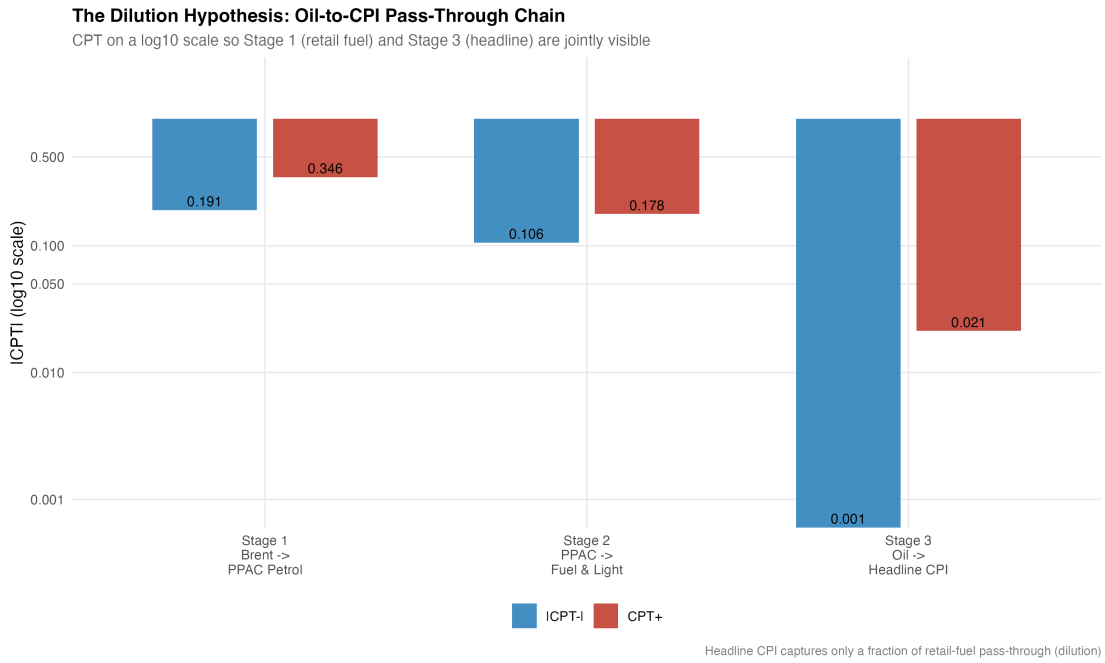

Figure 7: Attenuation across layers. Cumulative positive pass-through (CPT^+) by stage, from rupee oil through retail petrol, wholesale and consumer fuel-sensitive subindices, and the two headline aggregates. The headline-CPI bar is the smallest and is not statistically distinguishable from zero in the underlying ADL.

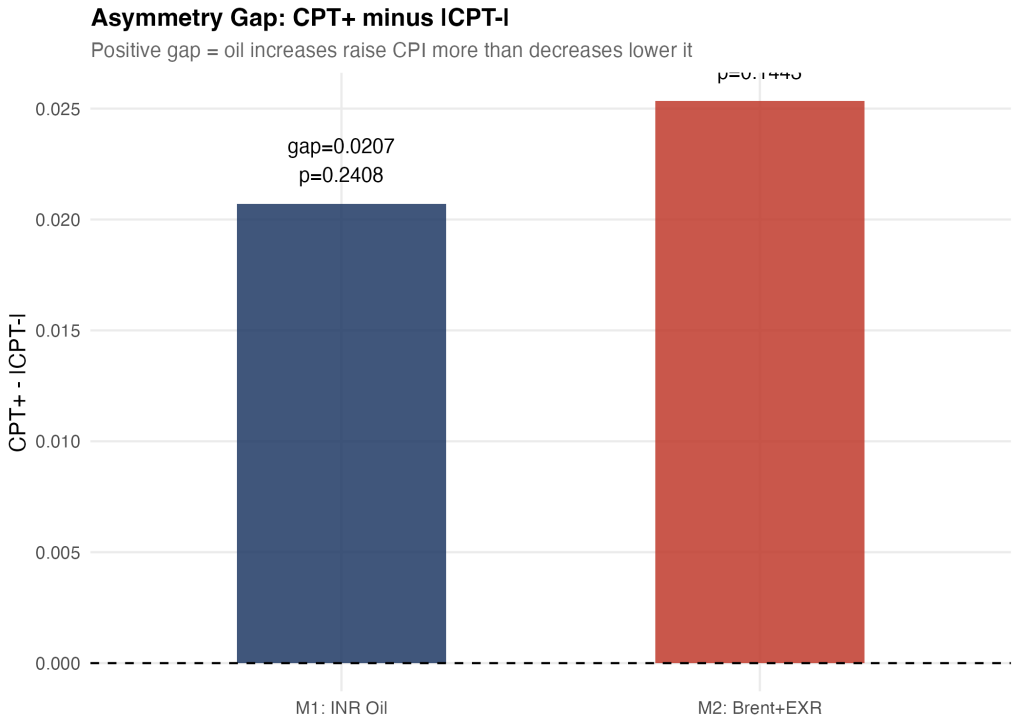

Figure 8: Asymmetry gap (CPT^+ minus CPT^-) by layer with HAC confidence bands. The gap is largest and signed positive at the retail-petrol stage, consistent with the conventional rockets-and-feathers reading. Both upstream-wholesale and headline-CPI gaps straddle zero.

6 Robustness and supplementary evidence

Three robustness exercises support the main reading.

First, the Brent + EXR decomposition. Replacing rupee oil with separate asymmetric Brent components and a contemporaneous EXR control yields $\text{CPT}^{+} = 0.031$ ($p = 0.023$) and $\text{CPT}^{-} = 0.037$ ($p = 0.001$) on the headline WPI sample. Adjusted R^2 rises to 0.42 and the symmetry test is not rejected ($p = 0.70$). The decomposition does not move the substantive conclusion, which suggests that the rupee oil aggregate used in the main specification is not masking offsetting movements in price and exchange rate components.

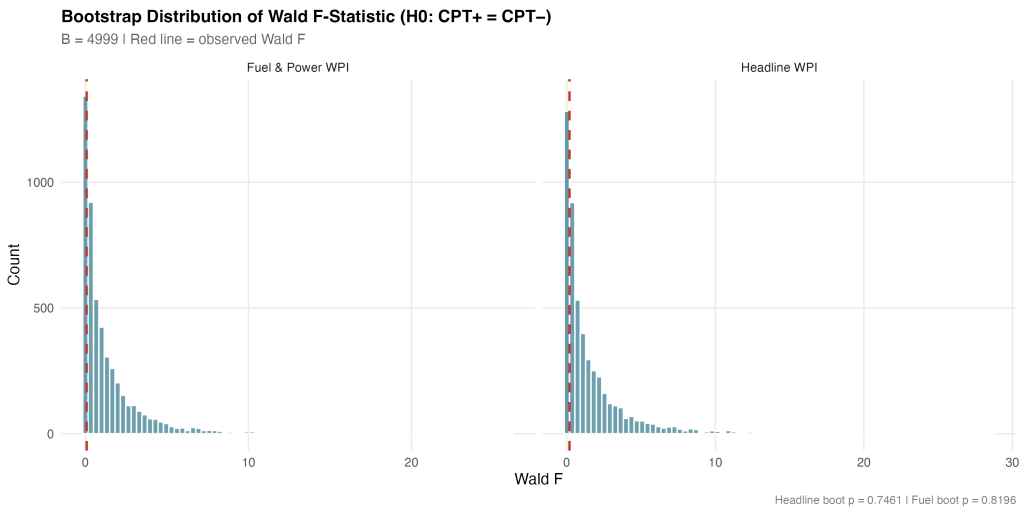

Figure 9: Bootstrap distribution of the symmetry Wald statistic for the headline WPI ADL. Restricted-residual circular block bootstrap, $B = 4{,}999$. The observed Wald statistic sits near the centre of the simulated distribution, which gives a bootstrap symmetry p-value of 0.746.

Second, the bootstrap symmetry tests. For both the headline WPI ADL and the WPI Fuel and Power ADL we resample residuals using a restricted-residual circular block bootstrap with 4,999 replications. The bootstrap symmetry p-values are 0.746 for headline WPI and 0.820 for fuel and power. There were no bootstrap failures. The asymptotic Wald conclusions hold up in finite samples on the wholesale models. The headline CPI bootstrap p-value is 0.50, also consistent with the asymptotic test.

Third, predictive precedence. Granger tests at four lags reject the null of no Granger causation from rupee oil to headline WPI ($F = 7.43$, HAC $p < 0.001$), from rupee oil to WPI Fuel and Power ($F = 15.22$, HAC $p < 0.001$), and from Brent in USD to headline WPI ($F = 7.38$, HAC $p < 0.001$). The reverse direction is not Granger-significant for headline WPI ($p = 0.43$). The exchange rate Granger-causes WPI at the 10% margin ($p = 0.05$). We treat these statistics as predictive precedence rather than as structural identification, in the conservative reading we adopt throughout.

A few sensitivity exercises round out the picture. Excluding the April–September 2020 COVID window does not materially change any of the headline pass-through estimates. Lag-sensitivity exercises moving the oil lag length between $q = 4$ and $q = 8$ leave the qualitative ranking of CPT^{+} across layers intact. Winsorising the rupee oil change at the 1st and 99th percentiles trims the negative-side coefficient for the WPI Fuel and Power model slightly but does not alter the conclusion.

The appendix battery of NARDL bounds models in levels gives the long-run perspective. All four NARDL specifications reject the no-cointegration null at the 1% upper bound of Pesaran et al. (2001). Each specification has a negative and significant error-correction coefficient. The long-run asymmetry

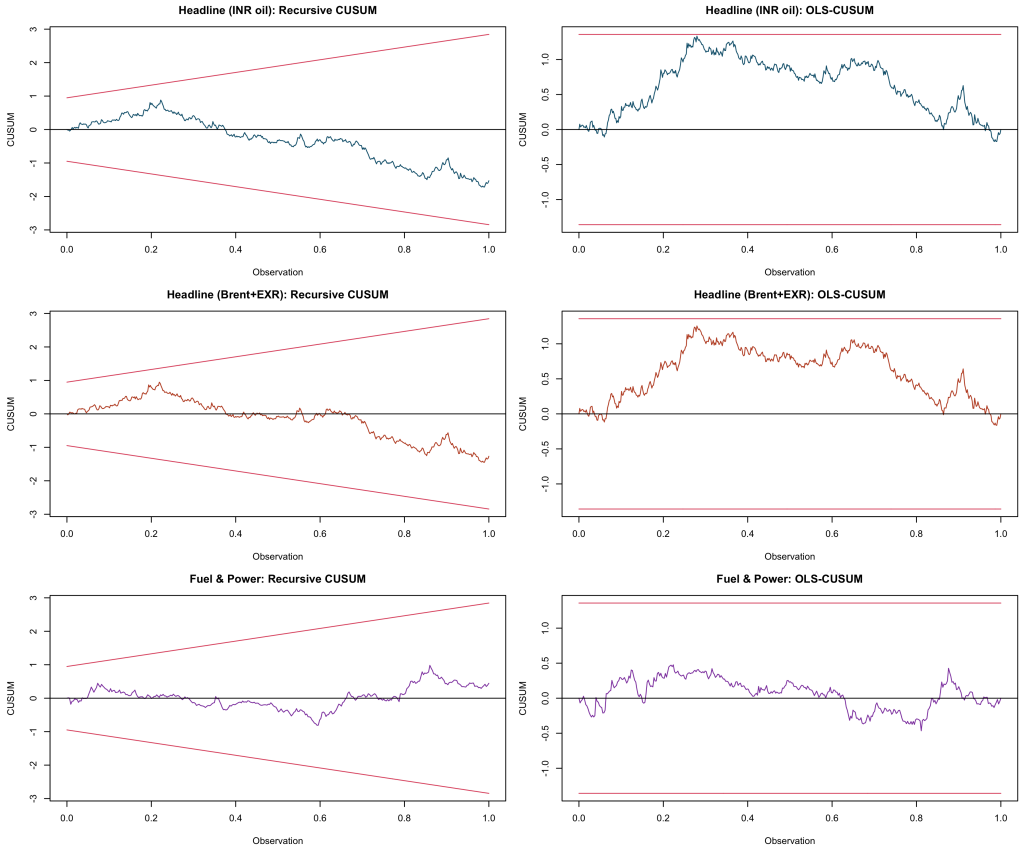

Figure 10: Recursive and OLS-CUSUM stability tests for the headline WPI ADL. Both test statistics stay inside their 5% bands across the 1983–2026 sample, consistent with parameter stability in the differenced specification despite the September 2013 break detected by the Bai–Perron test on the dependent variable.

tests of Shin et al. (2014) reject equality of long-run positive and negative multipliers in three of four specifications, with the headline INR-oil specification on the borderline ($p = 0.023$). The fuel and power specification has the largest ECT magnitude (-0.081, $p < 0.001$), which is consistent with the higher pass-through speed in that layer that the ADL also picks up. We read the appendix NARDL evidence as supplementary. It identifies asymmetry that the differenced ADL does not. But the long-run multiplier estimates are sensitive to the assumed lag structure and to the unit-root status of the level series, and we do not want the appendix tail to wag the main-text dog.

The diagnostics are not uniformly clean. The WPI Fuel and Power ADL fails the HAC RESET test at $p < 0.001$. We have already flagged this. It is the model in which we would invest least in the precise point estimate, and it is one of two models for which the structural-break literature suggests the differenced specification still leaves curvature on the table. The headline WPI ADL passes Breusch–Godfrey at $p = 0.84$, RESET at $p = 0.35$ and recursive CUSUM at $p = 0.39$. The headline CPI ADL passes its diagnostics. The bootstrap symmetry tests, taken together with the diagnostics, support the central empirical claim: the magnitudes of the upstream pass-through estimates are credible, the headline-CPI weakness is not an artefact, and the cross-layer attenuation pattern is the robust feature of the data.

7 Conclusion

This paper has analysed Indian oil pass-through as a transmission chain rather than as a single summary coefficient. The chain begins with the rupee cost of imported crude, moves through retail petrol pricing, enters the fuel-sensitive components of the wholesale and consumer price indices, and ends at the headline aggregates used in policy discussion. The transmission is clearly uneven. It is large at the pump (cumulative pass-through of about 0.35 for positive oil shocks), strong in the wholesale fuel layer (0.29), more moderate in the consumer fuel layer (0.18), and small to absent in the headline aggregates (0.03 for WPI, 0.02 and not statistically significant for CPI). The contrast between the pre-2010 and post-2010 estimates is consistent with a more market-linked pricing regime for transport fuels. Short-run asymmetry appears at the pump, but it is not the central or most robust result in the headline specifications. The supplementary NARDL evidence in the appendix points to long-run asymmetry, in line with parts of the recent literature, but the paper's main contribution lies in the integrated mapping of transmission and attenuation across layers.

Several policy implications follow. First, not all inflation measures respond to oil shocks in the same way. WPI is more informative about upstream cost transmission and imported energy-price pressures, while CPI is more informative about household-facing inflation. Treating them as interchangeable would blur an important distinction in the Indian case. Second, weak headline CPI pass-through should not be read as evidence that imported oil has ceased to matter. The mechanism results show that a substantial upstream shock is present, but that much of it is absorbed before it reaches the consumer aggregate. Third, institutional features matter for that absorption. The pre/post-2010 comparison and the retail-fuel evidence are both consistent with the view that pricing reforms, tax buffers, and fuel-specific channels shape how much of the external shock ultimately reaches households.

The paper should also be read with appropriate caution. The estimates are predictive rather than structural, and the analysis does not identify counterfactual policy effects. The stronger post-2010 coefficients are consistent with reform-era strengthening of pass-through, but the split-sample comparison should not be read as a standalone causal identification of deregulation. The retail-fuel sample begins only in August 2004, which limits the pre-deregulation window for the cleanest mechanism model. Diagnostic concerns on the fuel and power equation should be kept in mind when interpreting that magnitude. The appendix NARDL battery adds a useful nonlinear levels-based perspective, but it remains supplementary and depends on assumptions about the level-series properties that the KPSS evidence advises treating carefully.

Two extensions follow naturally from the present design. One is a finer disaggregation of the consumer basket. In this paper, the CPI Fuel and Light bridge equation is treated as supporting evidence because of its shorter sample, but several CPI groups such as housing, transport, and miscellaneous categories embed indirect oil exposure that the headline series conceals. A more harmonised long sample would make it possible to separate mechanical weight effects from broader transmission through related consumption items. The second extension is a state-space approach with time-varying pass-through coefficients, which would allow the reform-era strengthening to appear as a gradual shift rather than as a discrete subsample break. Both are left for future research.

The central message is straightforward. The relevant policy question is not whether WPI or CPI is the single correct inflation measure for studying oil shocks. It is where in the price system the shock survives and where it weakens. The evidence presented here suggests that the shock is visible in wholesale and

fuel-sensitive prices, but largely attenuates before it reaches headline consumer inflation. Making that layered pattern explicit is the paper's main contribution.

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Appendix A. Supplementary nonlinear long-run evidence

The NARDL bounds battery reported in this appendix follows Pesaran et al. (2001) for the bounds test and Shin et al. (2014) for the asymmetric long-run decomposition. Models are estimated with case 3 (unrestricted intercept, no trend) and AIC lag selection up to four lags. All specifications cover April 1997 to March 2025. Table 7 summarises the bounds-F statistic, the error-correction coefficient and its p-value, and short- and long-run asymmetry p-values. Long-run multipliers are reported in the project output file `wpi/outputs/tables/table_08_nardl_long_run.csv`; we omit the levels coefficients here for space.

Table 7: Appendix NARDL bounds battery. April 1997 to March 2025.

Specification	N	Bounds-F	ECT	ECT p	LR asym. p
$\ln \mathrm{WPI} \sim \ln \mathrm{Brent}$	333	27.66	−0.019	0.005	0.001
$\ln \mathrm{WPI} \sim \ln \mathrm{Brent} \mid \ln \mathrm{EXR}$	333	21.40	−0.019	0.006	0.000
$\ln \mathrm{WPI} \sim \ln \mathrm{Oil}^{\mathrm{INR}}$	331	21.43	−0.017	0.011	0.023
$\ln \mathrm{Fuel\&Power} \sim \ln \mathrm{Oil}^{\mathrm{INR}}$	331	34.15	−0.081	0.000	0.000

Notes: NARDL with case 3 (unrestricted intercept, no trend), AIC lag selection up to four lags. Bounds test critical values from Pesaran et al. (2001); the 1% upper bound for case 3 with $k = 1$ is 7.84 and for $k = 2$ is 6.36. All four bounds-F statistics exceed the 1% upper bound. ECT is the true error-correction coefficient (sum of all y-level-lag coefficients) with a Newey–West Wald p-value. LR asymmetry p is the long-run Wald test on the equality of positive and negative cumulative multipliers in the spirit of Shin et al. (2014).

A short caveat on KPSS framing is appropriate. KPSS on $\Delta \ln \mathrm{WPI}$ returns $\hat{\eta} = 0.90$ against a 5% critical value of 0.46, while ADF and Phillips–Perron strongly reject I(1) (statistics of -12.22 and -14.39 respectively). We follow the standard reading that KPSS is known to over-reject in long samples that contain structural breaks, of which the September 2013 Bai–Perron break in headline inflation is one example. A Bai–Perron test on the rupee oil change finds zero breaks. We retain the NARDL specifications because the bounds test is robust to ambiguous integration order, but we report long-run multipliers with caveats and prefer the differenced ADL for the body of the paper.